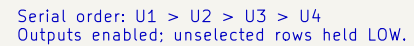
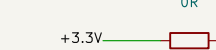


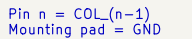
MCU serial stream > four latched drivers > 32-way row FFC



Clock / latch damping



32 columns > two 16:1 muxes > gain of 6 > filtered ADC inputs



R1/R2: 10k pulldown
Rf/Rg: 10k / 2k (G = 6)

YREF tracks ROW_VCC

RP2354A / clock / core supply / local bypass



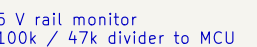
C20
15pf

MCU 3.3 V bypass

100nF

Harness or USB power > diode OR > local 3.3 V buck

Harness or USB power > diode OR > local 3.3 V buck



MCU analog supply filter

ROW_VCC link / VREF filter
see sections 01 and 02.
All returns use GND.

Half-duplex data pair and dedicated receive-only sync pair

Half-duplex data pair and dedicated receive-only sync pair



board address straps: ADDR0 / ADDR1 / ADDR2
losing a strap connects that input to GND.

tatus indicator

USB-C data / configuration straps / SWD / BOOTSEL

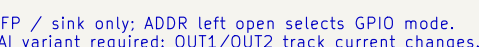
USB-C data / configuration straps / SWD / BOOTSEL



BOOTSEL: series resistor limits the flash-CS s

Computer port current detection > firmware authorization > current limit > reverse barrier > +5V_BUS

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Reading the sheet

Matching net labels connect between sections. A dot marks a branch; an X marks an unused pin. Bypass capacitors are grouped by supply. USB attach current and the complete eight-board startup / operating budget require measurements; schematic checks cannot establish them.

Sensor path: ROW_n > mat element > COL_n > SENSE_A/B > AMP_A/B > ADC_A/B

3.3 V logic | USB-C data and protected chain power | SWD debug

Row drive / dual-bank sensing / RP2354A / RS-485

File: rev3.kicad_sch

Title: TaxelScan / 32 x 32 sensor interface	
Size: User	Date:

KiCad E.D.A. 10.0.5	
15	16